Examiner only

5. The international trade of ivory has been banned since 1989. However, as many as 50 000 African elephants are killed each year for their ivory tusks out of a population of less than 400 000.

Researchers created a map of genetic profiles of different elephant populations across Africa using dung samples containing DNA from epithelial cells.

Polymerase Chain Reaction (PCR) was carried out to amplify the DNA and after 40 cycles over a billion copies of the target sequence was produced.

(a) (i) Explain the following processes during the PCR: [3]

I. a single stranded DNA primer is added;

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II. the DNA is heated to 95°C at the start of a cycle;

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III. the DNA is then cooled to 50–60°C.

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(ii) Explain why a certain type of polymerase, called Taq polymerase, is necessary in the final extension stage of the cycle at 70°C. [1]

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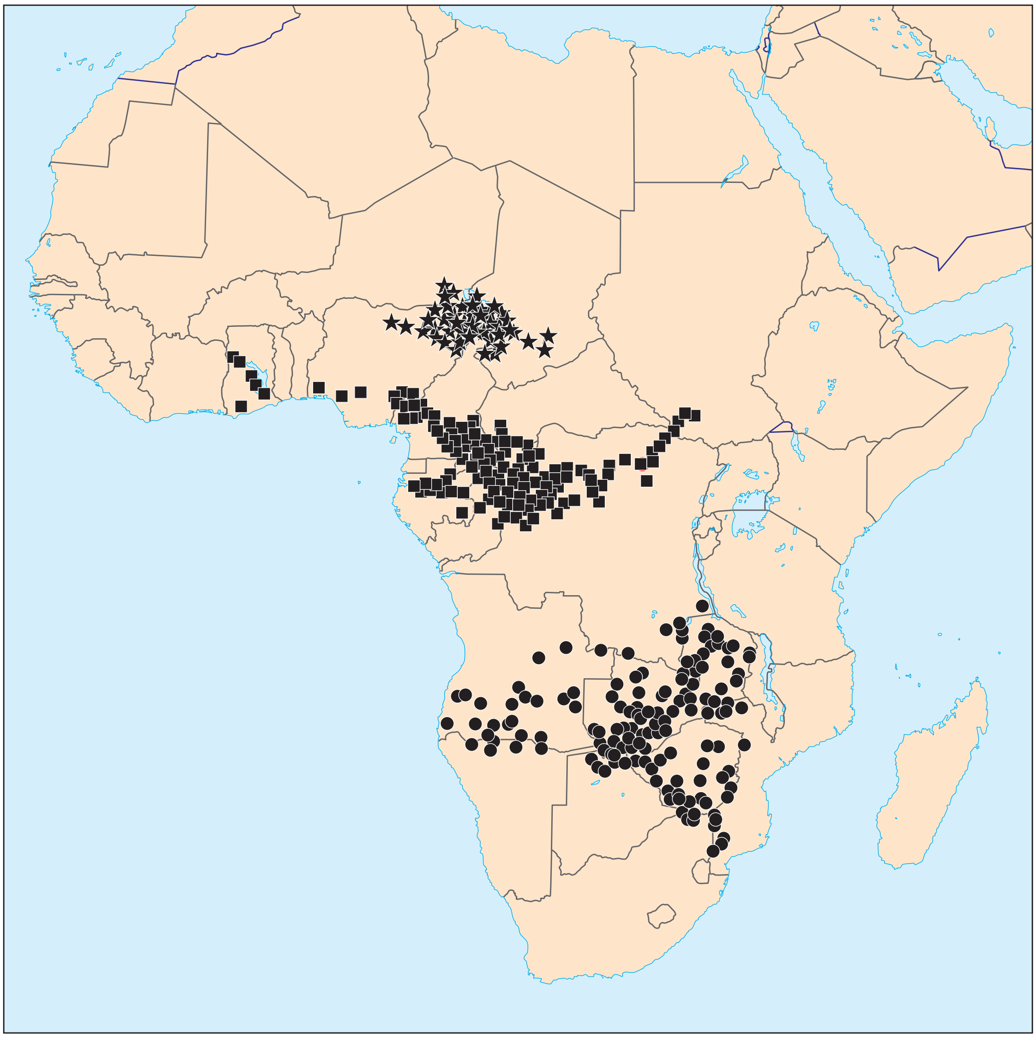
The researchers collected elephant dung samples from many locations to analyze the DNA. **Image 5** shows the distribution of these locations (★■●) . Each location was known to contain one population of elephants.

Image 5



(b) The different patterned dots (★■●) on **Image 5** indicate where closely matching genetic profiles were found by the researchers.
Explain the distribution of the genetic profiles of the different elephant populations shown on **Image 5**. [2]

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Most female African elephants have tusks, but typically about 6% of females in a population will never grow tusks.

However, in Gorongosa National Park in Mozambique:

- elephants with large tusks are targeted and killed by poachers for the illegal ivory trade;
- 33% of females between 10 and 20 years old do not have tusks;
- 50% of females over 20 years old do not have tusks.

(c) Using your knowledge of evolution, explain the high incidence of elephants without tusks in the Gorongosa elephant population. [3]

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(d) Suggest how the trend toward **increased lack of tusks** in a population with **heavy poaching** will affect African elephant population sizes in the future. Explain your answer. [2]

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(e) Ivory from elephant tusks contains DNA. Suggest how the data on the DNA profiles of populations could help with combatting poaching. [2]

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